AUTOSAR HLD Assistant – Complete Verification & Evaluation Summary

Test Document: Sample_AUTOSAR_HLD_Test_Document.pdf | LLM: qwen2.5:3b | Status: 15/15 PASS

1. Key Verification Parameters vs. Expected Results

Parameter / Metric	Expected	Actual Achieved	Status
Total Test Cases Executed	15	15	**PASS**
Tests Passed	15	15	**PASS**
Pass Rate (%)	100%	100.0%	**PASS**
Software Components Extracted	5	5	**PASS**
Interfaces Extracted	5	5	**PASS**
Document Pages Processed	6	6	**PASS**
Intentional Inconsistencies Detected	3	3	**PASS**
Active Local LLM	qwen2.5:3b	qwen2.5:3b (Ollama)	**PASS**
Source Citation Grounding	Filename + Page number	Grounded [Source, Page]	**PASS**
Anti-Hallucination on Unsupported Queries	No fabricated answers	Explicit refusal (Zero Hallucination)	**PASS**

2. Complete 15 Test Cases Execution Matrix

Test ID	Module	Input / Action	Expected Output	Status
TC01	Dashboard	Open application	Dashboard displays project statistics	**PASS**
TC02	Upload Documents	Upload sample HLD PDF	PDF uploaded and indexed successfully	**PASS**
TC03	Upload Documents	Upload same PDF again	Duplicate detected	**PASS**
TC04	AI Document Assistant	List all software components.	5 components identified with citations	**PASS**
TC05	AI Document Assistant	What are the dependencies of MotorControlSWC?	Correct dependencies with page references	**PASS**
TC06	AI Document Assistant	What happens when the brake request is invalid?	AllowedMotorTorqueNm = 0 Nm	**PASS**
TC07	AI Document Assistant	What is the CAN ID of the vehicle speed signal?	States information is not available	**PASS**
TC08	Architecture Analysis	Extract software components	5 components displayed	**PASS**
TC09	Architecture Analysis	Extract interfaces and signals	Documented interfaces and signals listed	**PASS**
TC10	Dependency Visualization	Generate dependency graph	5 components and documented relationships displayed	**PASS**
TC11	Consistency Checks	Run consistency analysis	Detects interface-name, timing, and torque-range mismatches	**PASS**
TC12	Export Reports	Export architecture analysis	Downloadable report generated	**PASS**
TC13	Export Reports	Export consistency findings	Report contains detected issues and references	**PASS**
TC14	AI Document Assistant	Summarize the system architecture.	Accurate summary with citations	**PASS**
TC15	Document Management	Delete uploaded document	Document and indexed data removed	**PASS**